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Roll No

ME-7003-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

OR and Supply Chain

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What is MRP? Explain MRP system and its functions giving suitable flow diagram. How this MRP is different from MRP-II?
b) Reduce the following game by Rule of Dominance and determine the game value and optimal strategies for both the players A and B:
2. a) Explain the role of decision making analysis in a business organization and also describe the steps involved in it.
b) Give short notes on the following:
 - i) Hurwicz criterion for decision making under uncertainty.
 - ii) Applications of Heuristic and Meta-Heuristic algorithms.
3. a) Discuss development of MRP to ERP system and SCM.
b) Discuss coordination and leadership issues in SCM process.

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4. a) What do you understand by queuing model? Write Kendall notation.
- b) Solve the following game graphically.

		B			
		I	II	III	IV
A	I	2	2	3	-2
	II	4	3	2	6

5. a) Derive the expression for economic order quantity.
- b) Explain ABC and VED analysis. What are their advantages?
6. a) What do you understand by travelling salesman problem? How is it solved?
- b) Deduce the "Little's law" formula. Give any one of its applications
7. a) Consider the following problem where unit transportation costs are given in Indian Rupees:

		Destinations				Availability
		J	K	L	M	
Factories	A	5	3	7	2	40
	B	6	4	2	8	70
	C	3	1	5	7	50
Requirements		75	20	50	30	

Due to shortage problems, some requirements may not be fulfilled. In such case, unfulfilled requirements may be levied penalties of Rs.2, Rs.5, Rs.3 and Rs.4 for the destinations J, K, L and M respectively. Determine the optimal allocation to minimize the transportation and penalty costs.

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- b) "Assignment Technique is a special case of Transportation Technique". Comment on this statement giving suitable example. Also enlist the industrial and non-industrial areas of application where assignment technique is used.
8. a) Write short notes on:
- i) Heuristic methods
 - ii) Metaheuristic methods
- b) Explain decision making and its types.
